

# Burglar Alarm System

A Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of

**B. Tech.**

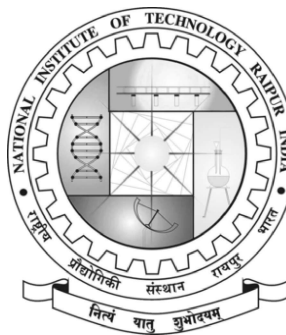
(Electronics and Communication Engineering)

By the Team (A' Batch, 3rd Sem, Session: **2025-26**)

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Under the guidance of

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# CERTIFICATE OF APPROVAL

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This is to certify that the project entitled “Burglar Alarm System”, submitted by the team comprising Deep Prakash (24116026), Deepak Jaiswal (24116027), Deepika Dwivedi (24116028), Dharmendra Bhaskar (24116029), Divyanshu Soni (24116030) has been carried out under my supervision in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology (B.Tech.) in Electronics and Communication Engineering at the National Institute of Technology, Raipur.

It is further certified that this work is original and has not been submitted elsewhere, in part or in full, for the award of any other degree or diploma.

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# ACKNOWLEDGEMENT

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We, the undersigned team members — Deep Prakash (24116026), Deepak Jaiswal (24116027), Deepika Dwivedi (24116028), Dharmendra Bhaskar (24116029), Divyanshu Soni (24116030) — would like to express our heartfelt gratitude and sincere thanks to our project guide, Dr. Kaushal Kumar Nigam, Assistant Professor, Department of Electronics and Communication Engineering, National Institute of Technology, Raipur, for his continuous guidance, valuable suggestions, and constant encouragement throughout the course of our project work. His expert insights and motivating supervision have been instrumental in the successful completion of our project, “Burglar Alarm System”.

We also extend our sincere thanks to all the Lab Assistants and Ph.D. Scholars, Department of Electronics and Communication Engineering, National Institute of Technology, Raipur, for providing the necessary facilities and academic support to carry out this work.

We are grateful to the institute for offering us this opportunity to apply our theoretical knowledge to a practical project and to enhance our technical and teamwork skills. Finally, we wish to acknowledge the cooperation, dedication, and mutual support of each team member, which made this work possible and fulfilling.

# ABSTRACT

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The Burglar Alarm System presented in this project is a cost-effective and reliable security circuit designed to detect unauthorized access through optical interruption. The system utilizes a laser–LDR (Light Dependent Resistor) configuration to sense intrusions.

Under normal operation, a continuous laser beam falls on the LDR, maintaining a constant resistance that keeps the circuit in an inactive state. When an intruder obstructs the laser path, the light intensity on the LDR decreases, resulting in a significant change in its resistance.

This variation alters the biasing conditions of the BC547 NPN transistor, causing it to switch on and activate a buzzer, which serves as an audible alert. The circuit is powered by a 9V battery, with an ON/OFF switch incorporated for operational control. A series combination of resistors ( $1\text{ k}\Omega + 1\text{ k}\Omega + 0.24\text{ k}\Omega$ ) ensures proper biasing and stable current flow within the circuit.

Owing to its simplicity, low power consumption, and easy implementation, this system is well-suited for basic home security applications and for educational demonstrations of electronic sensing and switching principles.

# Components Required

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1. LDR(Light Dependent Resistor)
2. Transistor – BC547
3. Buzzer
4. Switch Button(on/off)
5. Resistors ( $1\text{ k}\Omega + 1\text{ k}\Omega + 0.24\text{ k}\Omega$ )
6. Laser Light
7. DC Power Supply (9V)
8. Breadboard and Connecting Wires

## Working Principle

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The circuit operates on the principle of light-dependent resistance and transistor switching. A laser light is directed towards an LDR positioned at a fixed point. Under normal conditions, the continuous laser beam falling on the LDR keeps its resistance low, which in turn keeps the base voltage of the BC547 transistor insufficient to drive it into conduction. When the laser beam is interrupted by an obstacle, the resistance of the LDR increases sharply. This change in resistance alters the voltage divider network formed by the series resistors ( $1\text{ k}\Omega + 1\text{ k}\Omega + 0.24\text{ k}\Omega$ ) and the LDR, providing enough base voltage to turn ON the BC547 transistor. Once the transistor is switched ON, current flows through the buzzer, producing a sound that indicates an intrusion.

The circuit is powered by a 9V DC battery, and an ON/OFF switch is used to control power supply manually. The use of discrete components makes the design simple, robust, and suitable for demonstration or small-scale security applications.

# Objective

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The primary objective of this project is to design and implement a simple, low-cost burglar alarm circuit that can detect the interruption of a laser beam using basic electronic components. The system aims to:

- Provide an audible alert when any object or person obstructs the laser path.
- Demonstrate the use of LDRs and transistors as sensing and switching elements in security circuits.
- Develop a portable and battery-operated security prototype suitable for small-scale applications such as home or laboratory safety.
- Encourage understanding of basic electronic concepts including biasing, current control, and optical sensing.

# Circuit Diagram

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